

Program-2: Multi-Stage Dockerfile

Folder Structure

```
program-2/  
  |—— Dockerfile  
  |—— package.json  
  |—— src/  
      |—— index.js
```

STEP 1: Write src/index.js

Open the file:

nano src/index.js

```
GNU nano 7.2 index.js  
const express = require('express');  
const app = express();  
const PORT = 3000;  
  
app.get('/', (req, res) => {  
  res.send('Hello from multi-stage Docker!');  
});  
  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

STEP 2: Create package.json

nano package.json

```

GNU nano 7.2 package.json
{
  "name": "program-2",
  "version": "1.0.0",
  "description": "A Node.js app with multi-stage Dockerfile",
  "main": "dist/index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "mkdir -p dist && cp -r src/* dist/"
  },
  "author": "Deepika K",
  "license": "MIT",
  "dependencies": {
    "express": "^4.18.2"
  }
}

```

STEP 3: Install Node Modules

npm install

STEP 4: Create Multi-Stage Dockerfile

nano Dockerfile

```

GNU nano 7.2 Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./
RUN npm install

COPY . .
RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD ["node", "dist/index.js"]

```

STEP 5: Build Docker Image

docker build -t program-2 .

STEP 6: Run Docker Container

docker run -it -p 3000:3000 program-2

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program
[+] Building 48.2s (16/16) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 427B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:658d8f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448
=> => resolve docker.io/library/node:20-alpine@sha256:658d8f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448
=> sha256:8d06ba6946d1f299d8f962e37906c77086df33d47050430a57f7893ec35af697 42.78MB / 42.78MB
=> sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3 1.26MB / 1.26MB
=> sha256:658d8f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448 7.67kB / 7.67kB
=> sha256:fcbb8f7d018707c656a4da2eea8a15f2893d2258093fea9ff2ea5ea1c8a82112 1.72kB / 1.72kB
=> sha256:e80b0510ba947015cacddca3d23dc761e399971c074a1bf32eea7e44510524 6.52kB / 6.52kB
=> sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB
=> extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612
=> sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a52 443B / 443B
=> extracting sha256:8d06ba6946d1f299d8f962e37906c77086df33d47050430a57f7893ec35af697
=> extracting sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3
=> extracting sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a52
=> [internal] load build context
=> => transferring context: 2.33MB
=> [builder 2/6] WORKDIR /app
=> [builder 3/6] COPY package.json package-lock.json ./
=> [builder 4/6] RUN npm install
=> [builder 5/6] COPY . .
=> [builder 6/6] RUN npm run build
=> [stage-1 3/6] COPY --from=builder /app/package.json ./
=> [stage-1 4/6] COPY --from=builder /app/package-lock.json ./
=> [stage-1 5/6] COPY --from=builder /app/dist ./dist
=> [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules
=> exporting image
=> => exporting layers
=> => writing image sha256:ebbd0987d4777429dd61197e7908875cf699d32cc1efb963cbdf2377a0f7671d
=> => naming to docker.io/library/program-2

```

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program2$ docker images
REPOSITORY          TAG       IMAGE ID       CREATED        SIZE
program-2           latest    ebbd0987d477  15 seconds ago 137MB
flask-docker-app    latest    a1cf378133ca  19 hours ago   133MB
program4-app        latest    e1010a5e5baf  20 hours ago   194MB
program-6           latest    c9d43c3a9ca5  22 hours ago   912MB
yashwanth142003/program-6  latest    c9d43c3a9ca5  22 hours ago   912MB
ghcr.io/yashwanth142003/program-6  latest    c9d43c3a9ca5  22 hours ago   912MB
k1cbase/stable      v0.0.48  c6b5532e987b  3 months ago   1.31GB
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program2$ docker run -it -p 3000:3000 program-2
Server running on port 3000

```

STEP 7: Test in Browser

Open browser and go to:

<http://localhost:3000>

